

Course Code:	24CS303	Course Title:	COMPUTER ARCHITECTURE AND ORGANIZATION
Credits:	3	L – T – P	2-0-2
Pre-requisite	24CS331: Programming for Problem Solving using Python, Digital Logic Design		Common to AI-ML

Course objectives:

By the end of this course, students will be able to:

1. **Understand** the fundamental structure and operational principles of modern computer systems.
2. **Explore** instruction set architectures (ISA), addressing modes, and assembly-level programming.
3. **Analyze and design** core components of a processor, such as the ALU and control unit.
4. **Examine** memory organization, including cache and virtual memory systems.
5. **Understand and apply** pipelining and performance enhancement techniques in CPU design.
6. **Explore** I/O systems and recent trends in processor technologies such as multicore, RISC-V, and GPUs.

Teaching-Learning Process:

Suggested strategies that teachers may use to effectively achieve the course outcomes:

1. Interactive Lectures
2. Demonstrations and Simulations
3. Activity-Based Learning
4. Project based Learning
5. Experiential Learning
6. Flipped Classroom & Peer Learning

UNIT I – FUNDAMENTALS OF COMPUTER SYSTEMS AND ISAS	[6 hours]
Introduction to Computer Architecture: Von Neumann & Harvard Models , Evolution of ISAs: CISC, RISC, and the role of ARM, x86 – Components of a computer system , Instruction Set Architectures (ISA): Design Principles, Addressing Modes and Instruction Formats, Assembly Programming Basics (ARM/x86), Performance Metrics: CPI, MIPS, Amdahl's Law, Real-world ISA Case Study: ARM Cortex vs. Intel x86, simple assembly programs	
UNIT II – ARITHMETIC AND LOGIC UNIT	[6 hours]

Data Path Design: Components and Instruction Flow, ALU Design: Arithmetic and Logic Circuits Integer Arithmetic: Booth's Algorithm, Overflow Detection Floating Point Arithmetic: IEEE 754, Microinstruction Format and Execution , FSM-based Control Design , Introduction to Verilog/VHDL

UNIT III – MEMORY SYSTEM ARCHITECTURE	[9hours]
Memory Hierarchy Overview: Registers to Secondary Storage, RAM Types: SRAM, DRAM, Flash – Industry Applications, Cache Memory: Structure and Mapping Techniques, Cache Write Policies and Replacement Algorithms, Multilevel Caches and Performance Tuning, Virtual Memory: Paging, TLB, and Address Translation , Memory Management in OS (Real-world Focus: Linux)	

UNIT IV – PROCESSOR PERFORMANCE AND PIPELINING	[9 hours]
Instruction Pipelining: Basics and Stages, Pipeline Hazards: Data, Structural, Hazard Mitigation: Forwarding, Stalling, Branch Prediction, Superscalar and VLIW Architectures, Out-of-Order Execution and Speculative Execution, Introduction to Multicore and SMT (Simultaneous Multithreading), Modern Pipeline Case Study: Apple M-Series, AMD Ryzen	

UNIT V – I/O SYSTEMS AND FUTURE TRENDS	[9 hours]
I/O Interface Basics: Programmed I/O, Interrupts, DMA, I/O Bus Systems: PCIe, USB, NVMe (Industry Focus), Peripheral Integration in Embedded Systems, Device Drivers and Low-Level System Access, Overview of Embedded and IoT Architectures, RISC-V Architecture Overview and Applications, Introduction to GPU Architecture and Parallelism (CUDA)	

Course Outcomes:

On Completion of this course the students will be able to

CO	COURSE OUTCOME	Knowledge Level
CO1	Apply the concepts of instruction formats and addressing modes to write and analyze simple assembly-level programs.	K3
CO2	Apply arithmetic algorithms and control logic to design and simulate basic ALU and control units.	K3
CO3	Apply memory hierarchy principles to improve system performance using cache and virtual memory techniques.	K3
CO4	Apply pipelining techniques to improve instruction throughput and analyze CPU performance.	K3
CO5	Apply input/output techniques and explore modern processor features for efficient system integration	K3

COs and POs Mapping:

COs	Pos											
	1	2	3	4	5	6	7	8	9	10	11	12
CO1	3	2	1	1	1	-	-	-	2	-	-	1
CO2	3	2	1	1	-	-	-	-	2	-	-	1
CO3	3	2	1	-	-	-	-	-	2	-	-	1
CO4	3	2	1	1	-	-	-	-	2	-	-	1
CO5	3	2	1	1	1	-	-	-	2	-	-	1

Level 3- Highly Mapped, Level 2- Moderately Mapped, Level 1- Low Mapped, Level 0- Not Mapped

Scheme of Evaluation:

Component	Type of assessment	Max Marks	Reduced Marks	Total	Final marks
Continuous Internal Examination (CIE) – Theory	CIE – I	100	50	100	25
	CIE – II	100			
	MCQ	20	10		
	Skill Assessment - I	40	40		
	Skill Assessment - II	40			
Continuous Internal Examination (CIE) – Laboratory	Continuous Assessment	75	75	100	25
	Model Lab Exam	25	25		
End Semester Examination (ESE)	Theory Exam	100	35	50	50
	Lab Exam	100	15		
Total				100	

Assessment Pattern:

Bloom's Category	Continuous Assessment Tests		Terminal Examination
	1	2	
Remember	20	20	20
Understand	40	40	40
Apply	40	40	40
Analyze	0	0	0
Evaluate	0	0	0
Create	0	0	0

End semester Examination: (QP PATTERN)

- Each unit consists of two 2 marks questions and one 16 marks question (either or).
- All the fifteen questions have to be answered.

TEXT BOOKS:

1. David A. Patterson and John L. Hennessey, “Computer organization and design”,
2. William Stallings ,Computer Organization and Architecture, Pearson

REFERENCE BOOKS:

1. V.Carl Hamacher, Zvonko G. Varanescic and Safat G. Zaky, “Computer Organization”, VI the edition, Mc Graw- Hill Inc, 2012.
2. William Stallings “Computer Organization and Architecture”, Seventh Edition, Pearson Education, 2006.
3. Vincent P. Heuring, Harry F. Jordan, “Computer System Architecture”, Second Edition, Pearson Education, 2005.
4. Govindarajalu, “Computer Architecture and Organization, Design Principles and Applications”, first edition, Tata McGraw Hill, New Delhi, 2005.
5. John P. Hayes, “Computer Architecture and Organization”, Third Edition, Tata McGraw Hill, 1998.

Web Links and Video Lectures (E-Resources):

1. The evaluation of computer system:

<https://archive.nptel.ac.in/courses/106/105/106105163/>

2. Basic computer operations:

<https://archive.nptel.ac.in/courses/106/105/106105163/>

3/

3. Memory hierarchy Design

<https://archive.nptel.ac.in/courses/106/105/106105163/>

3/

4. Pipelining MIPS32 Data Path

<https://archive.nptel.ac.in/courses/106/105/106105163/>

5. RISC AND CISC Architecture

<https://archive.nptel.ac.in/courses/106/105/106105163/>

Activity-Based Learning /Practical-Based Learning:

1. <http://nptel.ac.in>
2. <https://swayam.gov.in>

Lab Exercises

1. Write and execute basic arithmetic and logic programs in Assembly (x86 or ARM).
2. Implement addressing modes using assembly code examples.
3. Design a 4-bit ALU using logic gates in Logisim (supporting AND, OR, ADD, SUB).

4. Implement Booth's Algorithm for signed multiplication in C or Assembly.
5. Simulate direct-mapped and associative cache mapping using C/Python or Excel.

6. Write a C program to demonstrate memory access latency with/without cache.
7. Simulate a 5-stage pipeline (Fetch, Decode, Execute, Memory, Write-back) for a given instruction set.
8. Identify and resolve pipeline hazards with forwarding and stalling mechanisms.
9. Simulate interrupt-driven I/O using C or assembly.
10. Design memory-mapped I/O interface for a simple peripheral (e.g., LED or switch) using Logisim.

Suggested Skill Activities:

1. Intel vs ARM architecture: A comparative analysis of two dominant architectures in mobile and desktop markets, with an exploration on why ARM has gained significant market share in mobile devices and why Intel still dominates PCs.
2. Design a simple ALU (4-bit or 8-bit) using Logisim that performs: ADD, SUB, AND, OR, XOR operations, Output flags: Zero, Carry.
3. Design a memory-mapped I/O system in Logisim to: Read switch input, Control LEDs or display output
4. **Mini Project:** To design, simulate, and demonstrate the working of a basic pipelined processor that supports arithmetic, logic, and memory instructions using a simplified instruction set.
5. **Mini Project:** Design a multi-bit ALU that performs arithmetic and logic operations. Supports ADD, SUB, AND, OR, XOR, NOT, Outputs Zero/Carry/Overflow flags
6. Visualize the decodg and execution of RISC-V instructions.

OBSERVE OPTIMIZE OUTSPREAD